EVALUACIÓN DE RESULTADOS

73.69 - LARGE LANGUAGE MODELS - 2026

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02. RAG

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01

EVALUACIÓN

¿Para qué evaluar los resultados?

Las métricas a utilizar dependen del caso de uso

DESAFÍOS

Data Leakage

El test set no debe estar “contaminado” con data del training set.
Los LLMs no deberían ser testeados en benchmarks públicos.

Reproducible

Al repetir el experimento con la misma configuración se deberían obtener resultados similares.

Coherencia y relevancia

El LLM debería producir resultados con sentido en relación al caso de uso.

02

RAG

EVALUACIÓN

RETRIEVER

¿Qué tan bueno es el contexto obtenido del vectorstore?

¿Es relevante?

¿Tiene ruido?

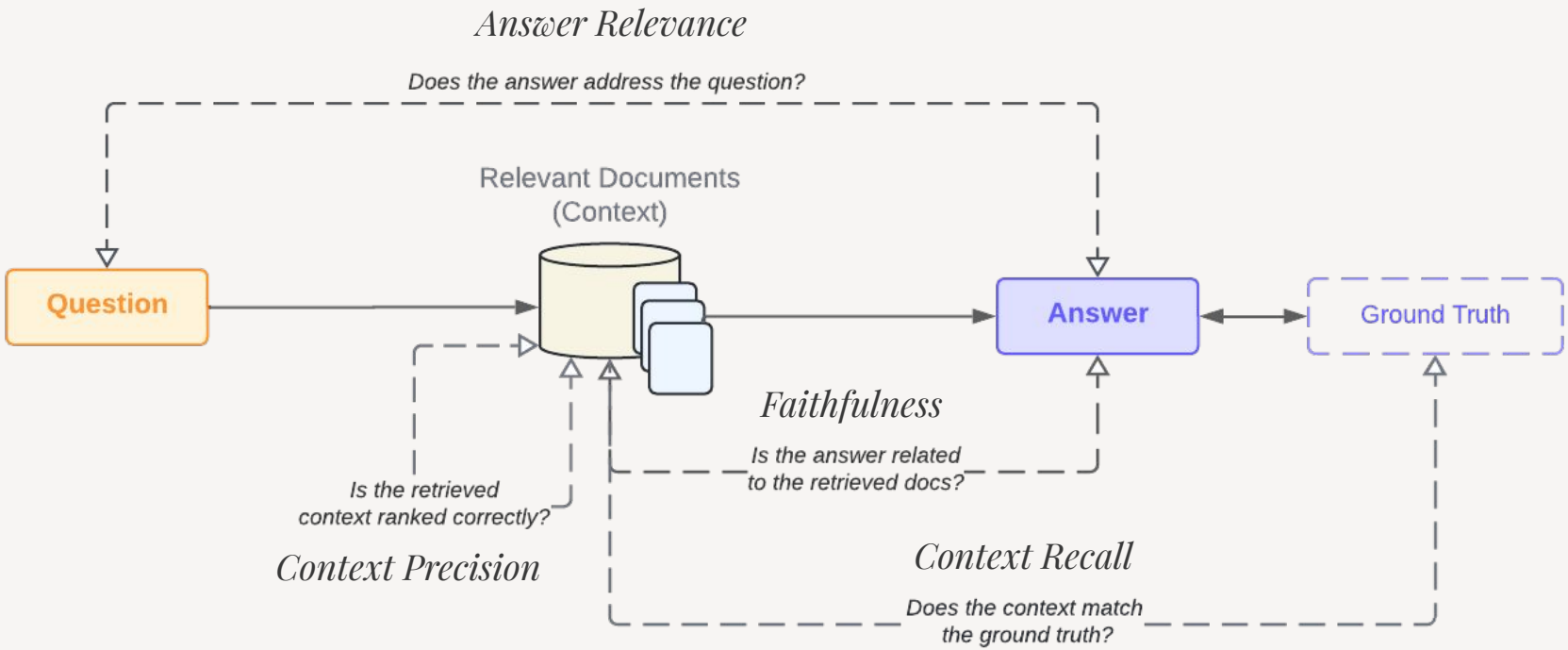
GENERATOR

¿Qué tan bueno es el contenido generado por el LLM?

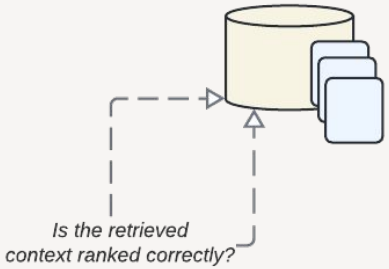
¿Es coherente con la query?

¿Incorpora el contexto?

RESUMEN


Answer Relevance
Does the answer address the question?
Relevant Documents
(Context)
Question
Answer
Ground Truth
Faithfulness
Is the answer related
to the retrieved docs?
Is the retrieved
context ranked correctly?
Context Precision
Context Recall
Does the context match
the ground truth?

RETRIEVAL METRICS



CONTEXT PRECISION

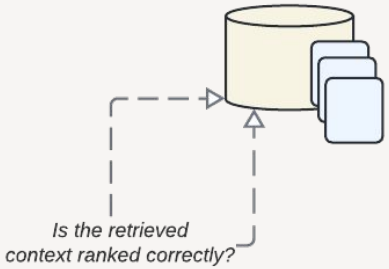
Determina si un chunk que se asemeja mucho a la GT está bien rankeado.

Dados k chunks y siendo $v_k \in \{0, 1\}$ un indicador de relevancia:

$$\text{Context Precision@K} = \frac{\sum_{k=1}^{K} (\text{Precision@k} \times v_k)}{\text{Total number of relevant items in the top } K \text{ results}}$$

$$\text{Precision@k} = \frac{\text{true positives@k}}{(\text{true positives@k} + \text{false positives@k})}$$

RETRIEVAL METRICS



CONTEXT PRECISION

Sea k=2 la cantidad de retrieved chunks

Prompt: “Where is France and what is its capital?

GT: France is in Western Europe and its capital is Paris

Chunks retrieved:
[“The country is also renowned for its wines and sophisticated cuisine. Lascaux’s ancient cave drawings, Lyon’s Roman theater”
“France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. Paris, its capital, is famed for its fashion houses, classical art museums including the Louvre and the Eiffel Tower”]

RETRIEVAL METRICS

CONTEXT PRECISION

$$\text{Context Precision@K} = \frac{\sum_{k=1}^{K} (\text{Precision@k} \times v_k)}{\text{Total number of relevant items in the top } K \text{ results}}$$

Etiquetamos los chunks con el indicador de relevancia v_k

Prompt: “Where is France and what is its capital?

GT: France is in Western Europe and its capital is Paris

Chunks retrieved:
[“The country is also renowned for its wines and sophisticated cuisine. Lascaux’s ancient cave drawings, Lyon’s Roman theater” $v_1 = 0$
“France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. Paris, its capital, is famed for its fashion houses, classical art museums including the Louvre and the Eiffel Tower”] $v_2 = 1$

RETRIEVAL METRICS

CONTEXT PRECISION

$$\text{Context Precision@K} = \frac{\sum_{k=1}^{K} (\text{Precision@k} \times v_k)}{\text{Total number of relevant items in the top } K \text{ results}}$$

Precision@1 = TP / (TP+FP) = 0 / (0+1) = 0

Precision@2 = TP / (TP+FP) = 1 / (1+1) = 0.5

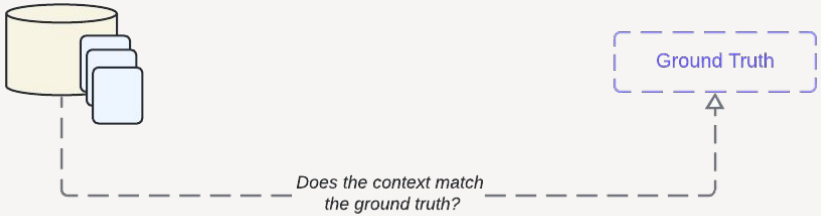
$$CP = \frac{0 * 0 + 0.5 * 1}{1}$$

Prompt: "Where is France and what is its capital?

GT: France is in Western Europe and its capital is Paris

Chunks retrieved:
["The country is also renowned for its wines and sophisticated cuisine. Lascaux's ancient cave drawings, Lyon's Roman theater" $v_1 = 0$
"France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. Paris, its capital, is famed for its fashion houses, classical art museums including the Louvre and the Eiffel Tower"] $v_2 = 1$

RETRIEVAL METRICS



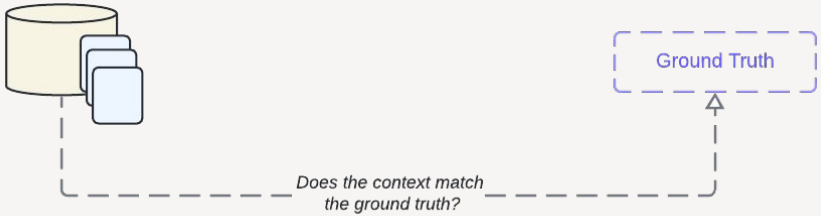
CONTEXT RECALL

Determina si todas las respuestas relevantes se encuentran en el contexto proporcionado

$$\text{context recall} = \frac{|\text{GT sentences that can be attributed to context}|}{|\text{Number of sentences in GT}|}$$

GT = ground truth

RETRIEVAL METRICS



CONTEXT RECALL

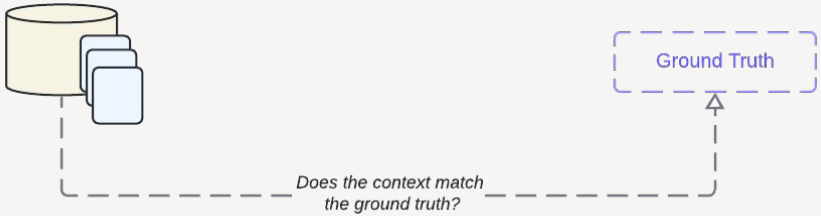
Determina si todas las respuestas relevantes se encuentran en el contexto proporcionado

Prompt: “Where is France and what is its capital?

GT: France is in Western Europe and its capital is Paris

Context: France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. The country is also renowned for its wines and sophisticated cuisine. Lascaux’s ancient cave drawings, Lyon’s Roman theater and the vast Palace of Versailles attest to its rich history

RETRIEVAL METRICS



CONTEXT RECALL

$$\text{context recall} = \frac{|\text{GT sentences that can be attributed to context}|}{|\text{Number of sentences in GT}|}$$

GT ST1 = True → sumo 1
GT ST2 = False → sumo 0 → CR = 1+0 / 2 = 0.5

Prompt: "Where is France and what is its capital?

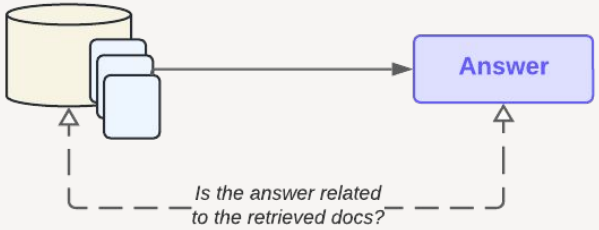
GT ST1 **GT ST2**

GT: France is in Western Europe and its capital is Paris

Se revisó el contexto y no estaba

Context: France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. The country is also renowned for its wines and sophisticated cuisine. Lascaux's ancient cave drawings, Lyon's Roman theater and the vast Palace of Versailles attest to its rich history

GENERATION METRICS

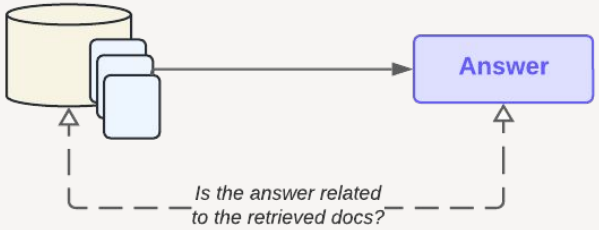


FAITHFULNESS

¿El LLM considera el contexto provisto? ¿Alucina?
A menor score, menos confío en la respuesta generada

$$\text{Faithfulness score} = \frac{|\text{Number of claims in the generated answer that can be inferred from given context}|}{|\text{Total number of claims in the generated answer}|}$$

GENERATION METRICS



FAITHFULNESS

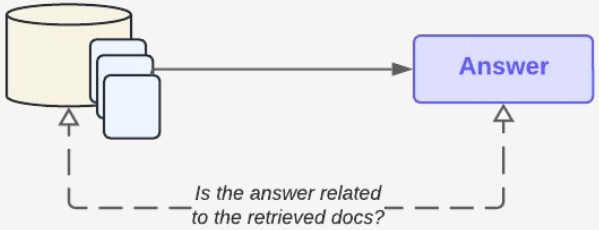
¿El LLM considera el contexto provisto? ¿Alucina?
A menor score, menos confío en la respuesta generada

Prompt: When was Attention Is All You Need published and by whom?

Context: Attention Is All You Need was published by Vaswani et al in 2017

Answer: Attention Is All You Need was published by Taylor Swift in 2017

GENERATION METRICS



FAITHFULNESS

$$\text{Faithfulness score} = \frac{|\text{Number of claims in the generated answer that can be inferred from given context}|}{|\text{Total number of claims in the generated answer}|}$$

¿El statement/claim puede ser inferido a partir del contexto?
GenAns ST 1 → False → 0
GenAns ST 2 → True → 1 → F = 1+0/2 = 0.5

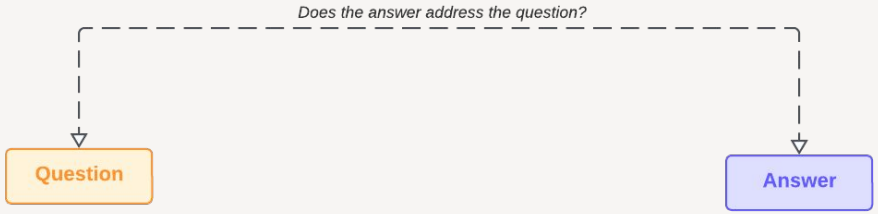
Prompt: When was Attention Is All You Need published and by whom?

Context: Attention Is All You Need was published by Vaswani et al in 2017

GenAns ST 1 **GenAns ST2**

Answer: Attention Is All You Need was published by Taylor Swift in 2017

GENERATION METRICS



ANSWER RELEVANCE

¿El LLM responde información irrelevante o relevante?

El score es menor cuando la respuesta es incompleta

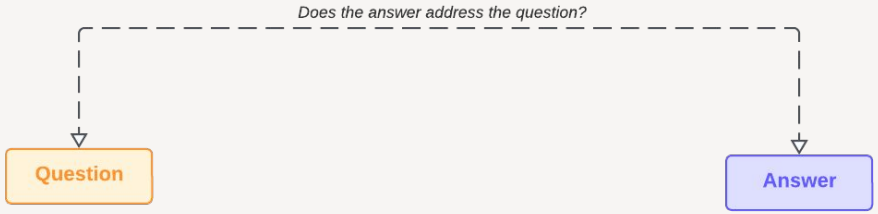
$$\text{answer relevancy} = \frac{1}{N}\sum_{i=1}^{N} cos(E_{g_i}, E_o)$$

E_{g_i} Embedding de la i ésima pregunta generada

E_o Embedding del prompt original

N Cantidad de preguntas generadas

GENERATION METRICS



ANSWER RELEVANCE

Usando otro módulo de LLM se generan múltiples preguntas similares, a partir de la respuesta (reverse engineering)
y se comparan los embeddings

Prompt: When was Attention Is All You Need published and by whom?

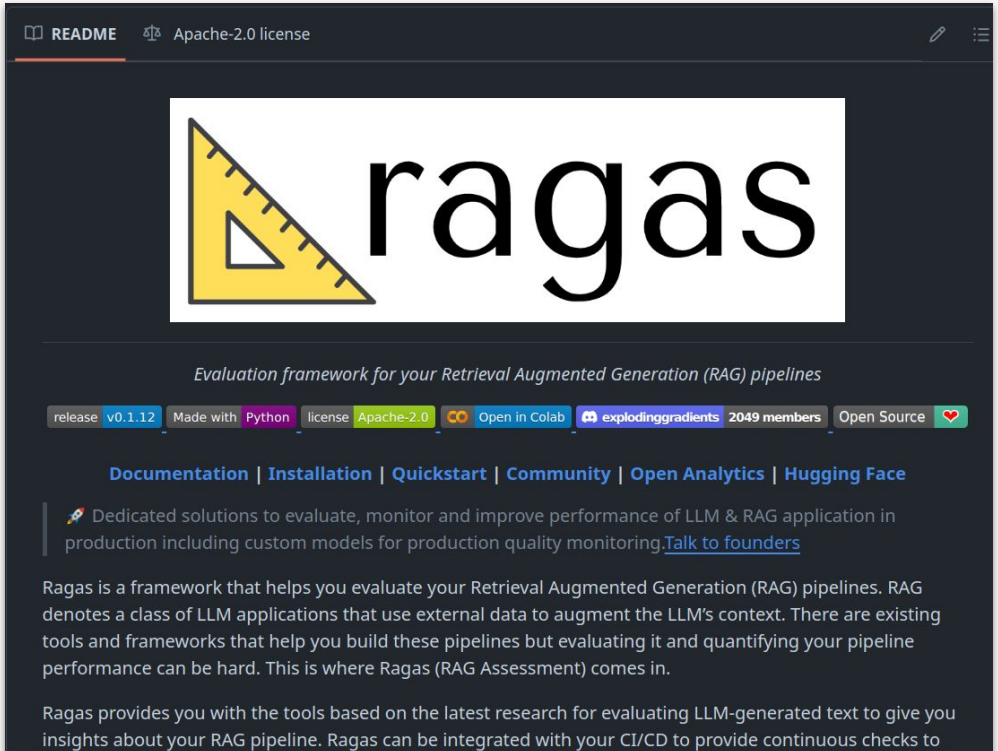
Question 1: “In what year was Attention Is All You Need published, and who were its authors?”

Question 2: “What is the publication date of Attention Is All You Need , and who are the authors?”

Question 3: “In which year was Attention Is All You Need paper presented, and who contributed to its creation?”

RAGAS

RAG Assessment Evaluation framework



DEMO



RAGAS.ipynb

¿Cómo evalúo mi modelo?

RESUMEN

QUALITY SCORES

- **Context Precision:** El contexto que me trajo el retrieval es correcto?
- **Context Recall:** Es el contexto lo suficientemente bueno para dar una respuesta?

- **Answer Relevance:** La respuesta efectivamente responde la pregunta?
- **Answer Faithfulness:** La respuesta está usando el contexto?

OTRAS MÉTRICAS

ABILITIES: Qué tan bien se puede adaptar? *Capacidad del modelo de:*

- **Noise Robustness:** manejar documentos relacionados pero que no tienen información sustancial
- **Negative Rejection:** no responder cuando no tiene la suficiente información
- **Information Integration:** sintetizar información de muchos docs

RANKING: Recommendation systems metrics, ej: MMR, NDCG (retriever)

COST: Tokens, API calls, infra

POSIBLES ACCIONABLES

GENERATOR METRICS	RETRIEVER METRICS: SCORE ALTO	RETRIEVER METRICS: SCORE BAJO
SCORE ALTO	Caso ideal	Revisar indexing params
SCORE BAJO	Prompt Engineering Model params	Revisar la arquitectura y la KB

03

OTRAS MÉTRICAS

Audios e imágenes

¿Cómo evalúo los resultados al utilizar audios?

WER

- Utilizada en ASR
- Porcentaje de palabras que el LLM predice incorrectamente en la transcripción

$$WER = \frac{S + D + I}{N}$$

Donde:

- N es el total de palabras en el texto de referencia
- S es la cantidad de sustituciones
- D es la cantidad de palabras que se deberían eliminar
- I es la cantidad de inserciones

¿Cómo evalúo los resultados de un modelo donde el output son imágenes?

Depende del caso de uso
Ejemplo: Structural Similarity Index (SSIM), Scale-Invariant Feature Transform (SIFT), comparación de histogramas, Classification, evaluación visual/contextual, evaluación por users

03

BIBLIOGRAFIA

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RAGAS Documentation https://docs.ragas.io/en/latest/concepts/metrics/index.html

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2024). ***Retrieval-Augmented Generation for Large Language Models: A Survey.*** Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University; Shanghai Key Laboratory of Data Science, School of Computer Science, Fudan University; College of Design and Innovation, Tongji University. URL https://arxiv.org/pdf/2312.10997

Evaluation metrics for ASR https://huggingface.co/learn/audio-course/chapter5/evaluation